PowerGrid EV Analytics: Data Cleaning and EDA Report (2022-24)

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1- Introduction:

This report analyses EV charging data that has three sheets (Chargers, Sessions and Stations). It covers data cleaning, exploratory data analysis and the key findings.

2- Data Overview:

The dataset has three sheets. The table below shows the number of rows and columns in each sheet and what it contains.

Sheet	Rows	Columns	Description
Chargers	199	5	Details of each charger: Charger ID, Station ID, Charger type, Max power and installation date
Sessions	294024	15	Each charging session: time, duration, energy, price, cost, payment method and user type
Stations	32	4	Details of each station: name, region, activation year and number of chargers

The data covers the years 2022 to 2024.

3- Data Cleaning:

The data was already in good condition. I checked all three sheets for missing values, errors, duplicates, inconsistent text and outliers, and found no problems. Data types were correct. I

removed the time part from the installation date and kept only the date. Column names are in lowercase with underscores.

4- Exploratory Data Analysis

Summary Statistics:

The table below shows the mean, median and mode of the numeric columns in the dataset. Stations and Charger sheets have only one useful numeric column.

Sheet	**Columns**	**Mean**	**Median**	**Mode**
Chargers	Max power kW	128.6	150	50
Sessions	Session duration	32.9	27	24
	Energy kwh	49.37	43.3	80
	Price per kwh	0.511	0.59	0.51
	Total cost	25.2	24.77	40.8
Stations	n charger	5.9	6	8

Distribution:

The table below shows the range and the most common value (mode) of each numeric column, based on the value distribution in Power BI.

Columns	**Min**	**Max**	Most Common Values
Max power kW	50	300	50
Session duration	15	58	24
Energy kwh	25kwh	80	80
Price per kwh	0.45	0.60	0.51
Total Cost	12.03	48	40.8

Categorical Analysis:

The table below shows the number of sessions in each category.

Payment Method	**Sessions**
App	98,267
RFID	97,863
Card	97,894

All groups have almost equal numbers of sessions, so no category dominates.

Relationships:

Scatter charts were used to check how the numeric columns in the Sessions sheet are related to each other.

Columns compared	**Relationship**	**What it shows**
Energy (kWh) vs Total Cost	Strong positive	Total Cost increases as Energy increases, because cost is calculated from energy and price. The fan shape appears because the price per kWh varies from 0.45 to 0.80.
Session Duration vs Energy (kWh)	Positive, three groups	Each charger type forms its own group: 50 kW (40-60 min, 21-45 kWh), 150 kW (20-30 min, 32.5-69.5 kWh), 300 kW (15-30 min, 49-80 kWh).
Price per kWh vs Energy (kWh)	No relationship	No relationship. Higher energy does not mean a higher price, so the data does not show demand-based pricing.

Trend Analysis:

Trends were checked using line charts, with the year on the X-axis and the count or total on the Y-axis. The table below shows how chargers, stations, sessions, energy and revenue changed from 2022 to 2024.

Measure	**2022**	**2023**	**2024**
Chargers installed	43	59	97
Stations activated	8	8	16
Sessions	40,640	86,132	167,252
Total energy (million kWh)	2.01	4.05	7.75
Total revenue (million)	1.03	2.59	4.55

All measures grew each year, and most of the growth came in 2024, when the number of chargers and stations expanded the most.

5- Anomalies and Questions:

One anomaly was found in the Energy (kWh) column. A large number of sessions are at 80 kWh, which is also the maximum value. This cluster appears only on 300 kW chargers. It may be a cap in the data or a battery limit of the vehicles. This should be confirmed with the data provider.

I have a Question: Are the 80 kWh values real readings or were higher values capped at 80?

6- Conclusion:

The EV charging dataset was in good condition. Missing values, errors and inconsistent text were not found, and the data types were correct. Most chargers are 50 kW, and the charger type decides how much energy a session delivers. Payment method and user type are spread almost equally across sessions. Chargers, stations, sessions, energy and revenue all grew from 2022 to 2024, with the biggest growth in 2024. The only anomaly is the large cluster at 80 kWh on 300 kW chargers, which should be confirmed with the data provider.